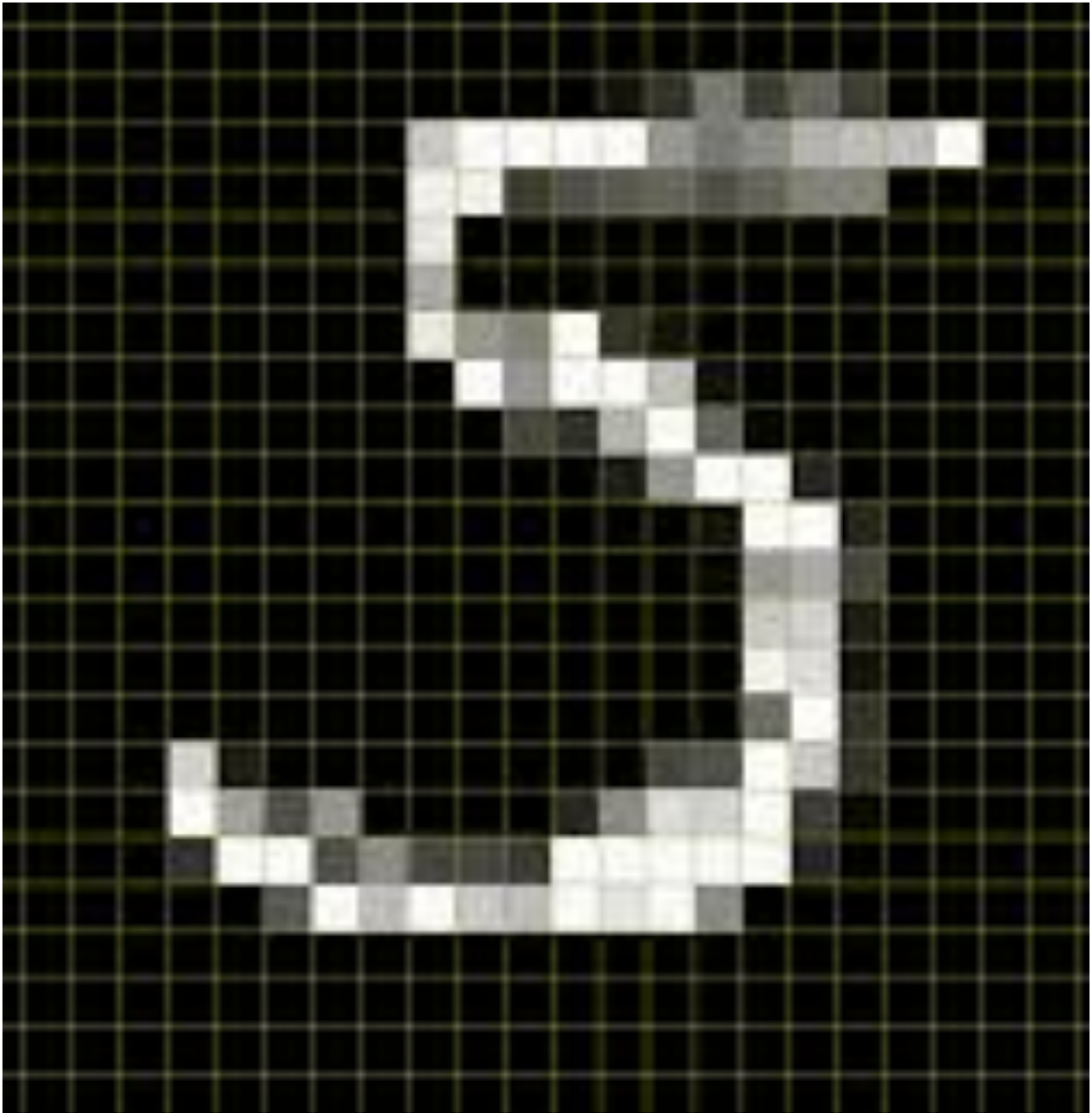
What does it mean to be an image?

Imagine an image as a scalar field $\phi(x)$ on $\mathbb{R}^2$.

$$\phi(x) =$$

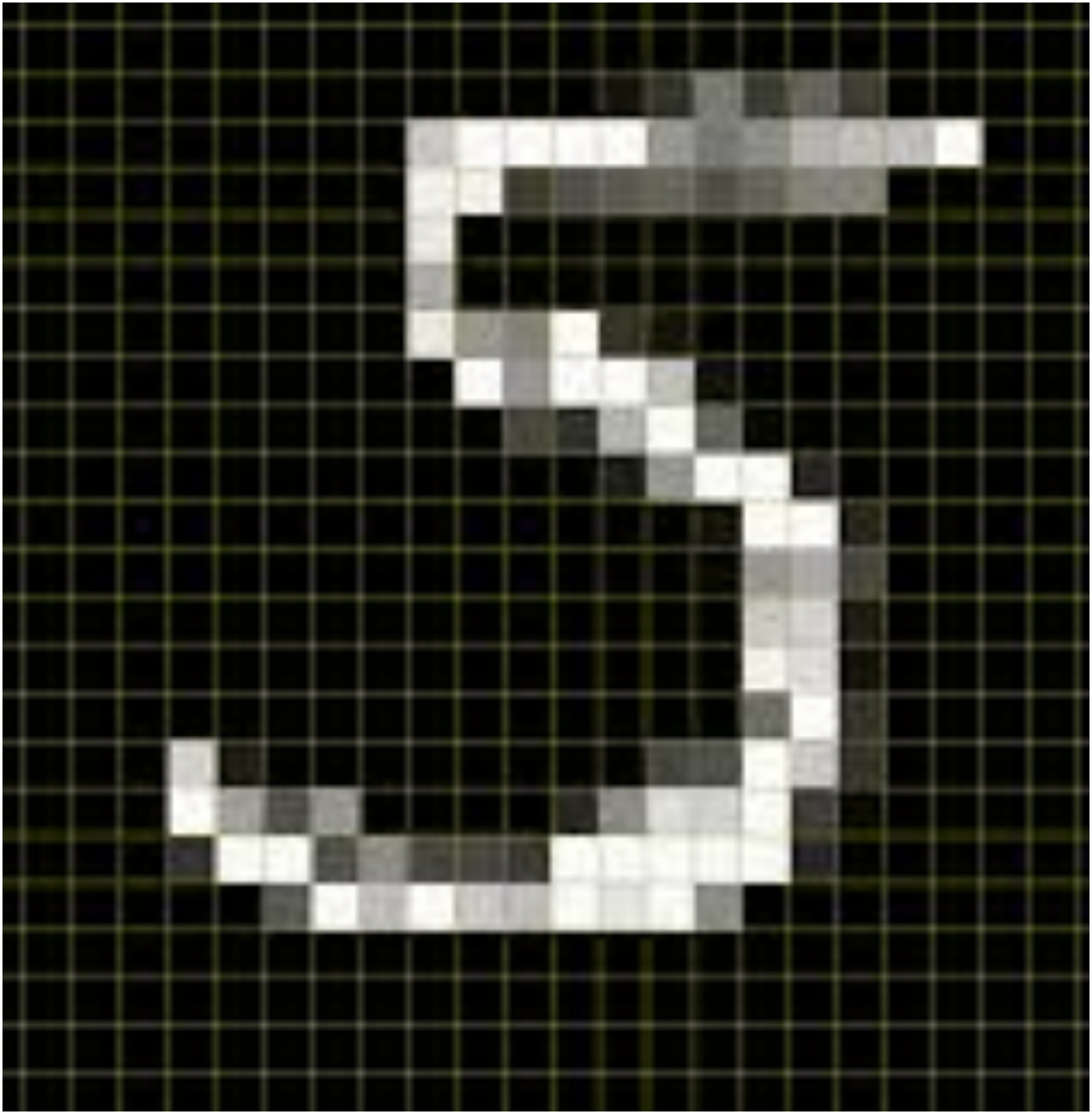


- **Translation**: if $\phi(x)$ is a "5", then so is $\phi(x+a)$.

- **Locality**: the pixels $\phi(x)$ and $\phi(y)$ should be considered "together" by the network if $|x-y|$ is small.

- This idea — that of **translational equivariance** — is encoded in machine learning in a **convolutional neural network,** where layers look like this:

$$\phi(x+a) =$$



$$W[\phi](x) = \int d^2\Delta \; K(\Delta)\phi(x + \Delta)$$

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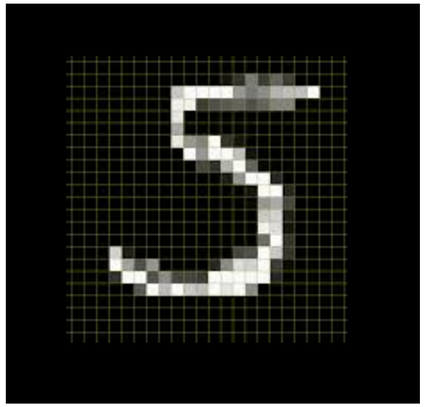
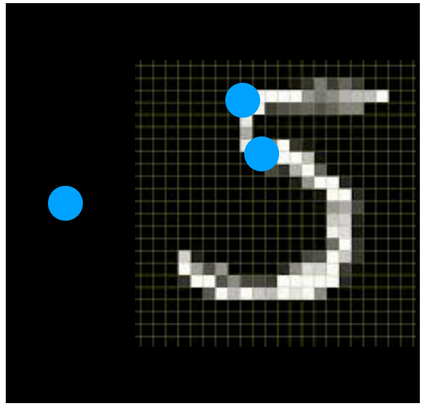


$\phi(x+a) =$



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Usefulness of translational equivariance

- This idea works **extremely well**. **ImageNet** is an image classification task with 14 million images, 20K categories.

ImageNet Classification with Deep Convolutional Neural Networks

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Imagenet classification with deep convolutional neural networks
A Krizhevsky, I Sutskever… - Advances in neural …, 2012 - proceedings.neurips.cc
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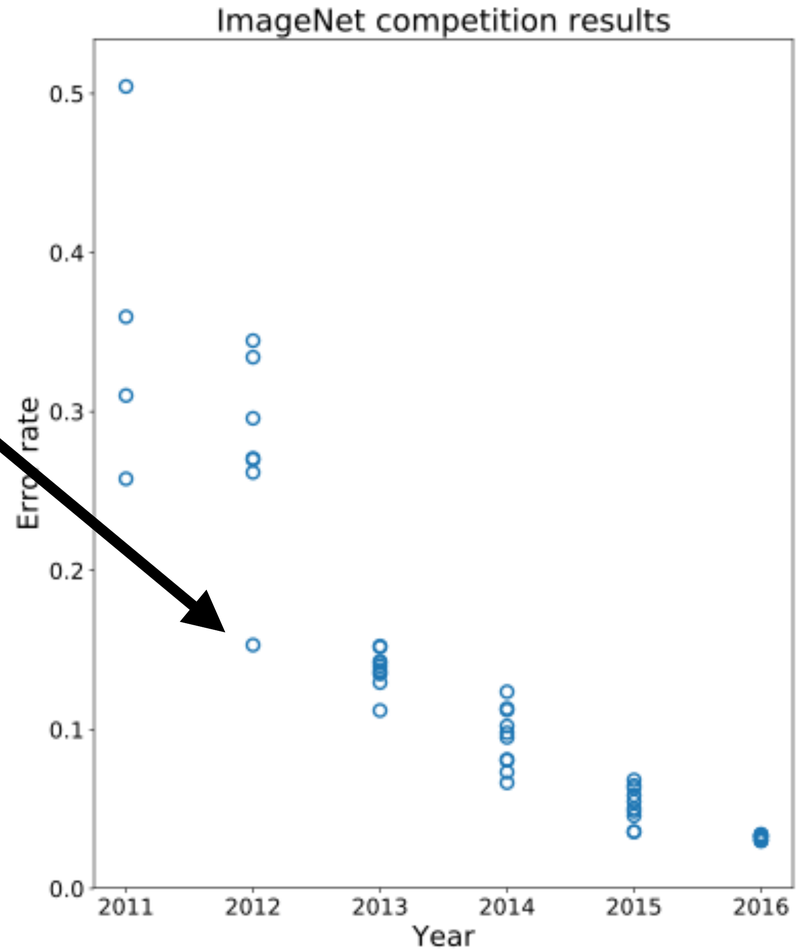

- This result helped start the deep learning revolution.